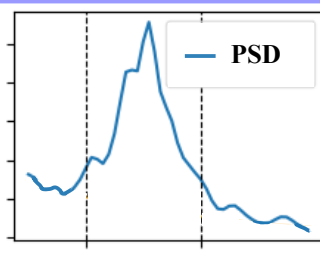


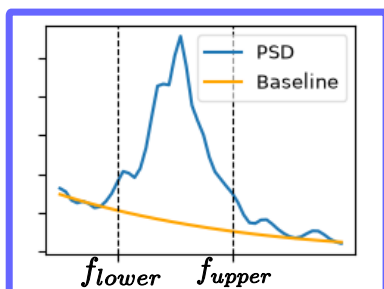
Step 0: PSD

- Project onto frequency axis to get PSD



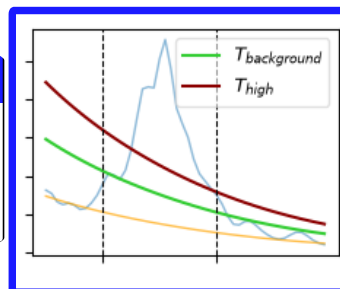
Step 1: Baseline

- Fit PSD bump with Whitaker baseline filter



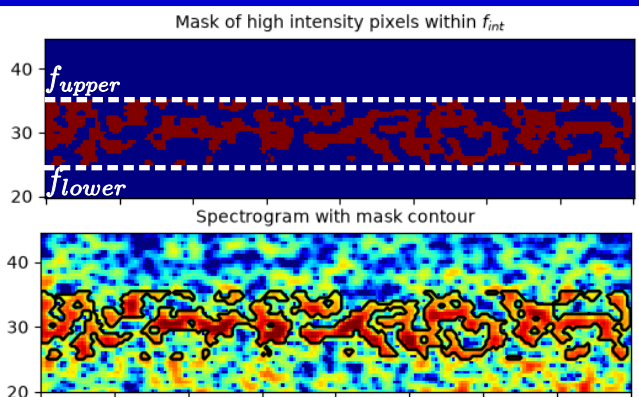
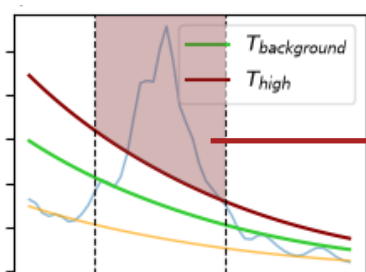
Step 2: Thresholds

- Multiply baseline by upper factor
- Multiply baseline by background factor



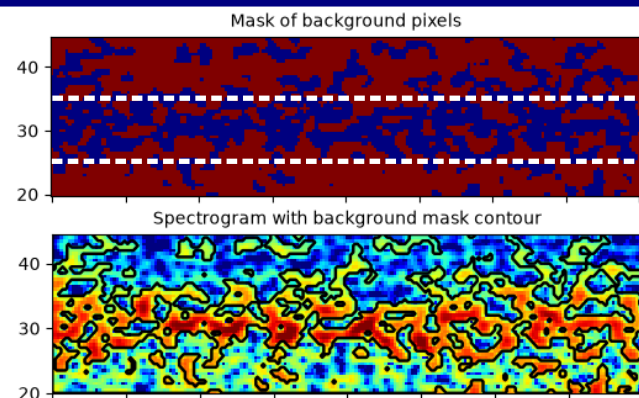
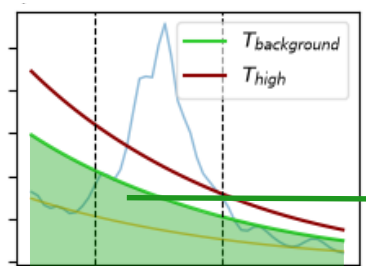
Step 3: high intensity pixel segmentation

- keep pixels above T_{high}
- Restrict only to pixels within $[f_{lower}, f_{upper}]$



Step 4: background pixels segmentation

- keep pixels below $T_{background}$

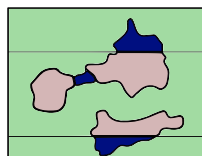


Step 5: mask refinement

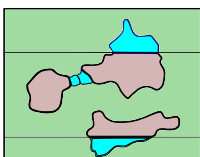
- Expand high intensity mask through watershed algorithm.

a. Original subdivision

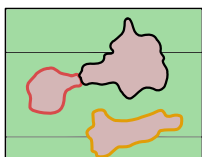
- $< T_{background}$
- $> T_{high}$
- $< T_{high}$ and $> T_{background}$



b. Expansion



c. Final Mask



Expanded connected parts

Mask after expansion

